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Strategic Development Paper

Since the start of the Russo-Ukrainian War on February 20, 2014, Russia has consistently used hybrid tactics—especially cyberattacks—as part of its aggression against Ukraine. These attacks have targeted critical infrastructure, including power stations, telecommunications, government and military networks, and transportation systems. The goal is clear: to weaken Ukraine's defenses and pave the way for Putin to reassert control and reclaim Ukraine as Russian territory. Given the scope of these actions, the United States and its allies must respond strategically to contain Russia's cyber aggression.

This memo outlines:

- A desired strategic end state for the cyber dimension of the conflict
- Three proposed strategic tenets to achieve the desired end state
- Supporting arguments drawn from relevant literature

The desired endstate is for Ukraine to obtain cyber soverignty, to support their overall goal to maintain control over their infrastrucutre, and ultinately defend themselves from Russian aggression. In order to achieve this, Ukraine must impement proper defense mechanisms, utlize threat-hunting strategies, and launch strateguc narrative campaings. These efforts will be led by the U.S., NATO, and allied partners acting in concert—a unified front that signals to adversaries that cyber aggression carries real consequences. Long term, we want to see tighter NATO cyber cooperation that goes beyond just coordination and turns into a real digital deterrent force—ready to respond when threats cross the line.

1 . Bolster Ukraine's Cyber Resilience

To achieve the desired endstate Ukraine will harden their critical infrastructure, such as power grids, telecoms, government, military, and transportation networks. The decentralization of control systems, will be implemented in order to reduce the risk of a single point of failure. Adding redundancy and maintaining cloud-based backups will help keep systems operational under attack.

A critical area of focus is the hardening of Industrial Control Systems (ICS). Russian cyberattacks have primarily targeted SCADA systems in Ukraine’s energy and water sectors. To counter this, Ukraine should segment ICS networks from traditional IT systems, promptly patch known vulnerabilities, and deploy intrusion detection systems (IDS) tailored to ICS environments to detect threats without disrupting operational technology. Adopting the Zero Trust Architecture model, where no user or device is trusted by default, can help harden systems. This would include:

- Strong identity verification (e.g., MFA, biometrics)

- Role-based access controls with least privilege enforcement and regular audits
- Identity and Access Management (IAM) platforms to separate duties and manage permissions
- Continuous monitoring through SIEM or SOAR tools to detect and respond to threats
- User and Entity Behavior Analytics (UEBA) to flag anomalies
- Endpoint Detection and Response (EDR) to identify malware or suspicious activity

2. Conduct proactive threat hunting and disrupt operations

Ukraine must formally invite USCYBERCOM into the country with legal authority to operate. This includes authorizing them to conduct hunt-forward operations, as outlined in a bilateral cyber cooperation agreement that clearly defines roles, rules of engagement, and data-sharing protocols. Once on the ground, the USCYBERCOM team would conduct deep network scans to identify malware and Russian implants and monitor network activity for indicators of compromise (IoCs).

The information gathered would be shared directly with Ukrainian cyber professionals to exchange tactics, strengthen defenses, and build long-term resilience. Similar operations have proven successful in countries like Montenegro, Estonia, and Lithuania.

In addition, cyber teams would be given the green light to preemptively disrupt Russian command-and-control servers, botnets, and malware infrastructure. Intelligence gathered from these missions would be shared with allied intelligence agencies and private-sector partners to help track Russian cyber activity worldwide.

All of this must operate under a clearly defined rules of engagement framework—designed to avoid escalation, remain below the threshold of armed conflict, but still apply enough pressure to disrupt and slow Russian attacks.

3. Implement Strategic communications and information warfare capabilities

Ukraine must develop real-time attribution using AI and human analysts to identify and publicly expose Russian cyberattacks within 48–72 hours, disrupting Moscow's ability to control the narrative. These attributions should be broadcast through Ukrainian media and official social channels.

Use government-backed platforms to trace disinformation flows—mapping bots, trolls, and fringe sites—while warning citizens about likely false narratives before they spread.

Coordinate messaging with NATO and Western media to reframe Russian cyberattacks on civilians as violations of international norms.

Fund digital campaigns and influencers to promote democratic resilience and weaken Russian influence at the societal level.

Conduct offensive cyber ops targeting Russian disinfo infrastructure—disabling C2 servers, doxxing troll networks, and corrupting their content databases.

Work with ISPs to identify botnets and spam farms, then flood Russian channels with noise, satire, and conflicting messages to degrade credibility.

Use cyber tools to disrupt Russian-language propaganda before it spreads through timestamp manipulation, translation errors, or redirecting feeds.

Partner with Meta, Google, and X to deploy AI tools that detect deepfakes and synthetic text before they go viral.

Literature review

Bolstering Ukraine's cyber resilience aligns with Schmidle's argument that a strong defense is essential for protecting against critical cyberattacks (Schmidle). This principle is applied in the first part of the plan by emphasizing the need to implement proper security controls. Schmidle also stresses that adhering to critical cybersecurity standards is key to denying adversaries easy access preventing Russia from exploiting Ukraine due to weak or improperly enforced controls (Schmidle).

Activating USCYBERCOM and allied partners offers several advantages. It can ease Ukraine's financial burden, as Freedman notes that international cooperation often leads nations to take on shared costs to punish nefarious actors and uphold standard norms (Freedman). Furthermore, *Joint Publication 3-12: Cyberspace Operations* highlights that offensive and defensive cyber missions require clear legal authorities and planning timelines (*Joint Publication 3-12*). Ukraine can address this by granting legal authority to the U.S. for hunt-forward operations. By leveraging this kind of cooperation, USCYBERCOM and its partners can proactively search for malicious activity on Ukrainian networks, helping to isolate and shut out states that provide safe havens for Russian botnets and troll farms (U.S. Department of Defense, *DIB Cybersecurity Strategy*). As Schmidle addresses, having both a capable offense and defense is crucial to defending American critical infrastructure against critical infrastructure attacks, and the same goes for this scenario.

Finally, leveraging information warfare is necessary to counter Russia's persistent disinformation campaigns. Russia uses information control as a strategic weapon to destabilize adversaries covertly (Bebber). This part of the strategy will focus on securing information flows since low-level cyber operations can build up and have long-term strategic consequences (Hoffman 131). To disrupt these efforts early, we must combine AI tools with human analysis to quickly identify and counter false narratives—neutralizing the adversary's offensive capabilities before they take hold (Schmidle).

The three strategic tenets outlined in this memo are essential to strengthening Ukraine's stability and resilience. To counter Russian hybrid warfare, Ukraine must invest in hardening its critical infrastructure and industrial control systems, proactively hunt for cyber threats, and implement robust strategic communications and information warfare capabilities. The steps prescribed here will help preserve Ukraine's cyber sovereignty, enhance its resilience, and damage the further success of Russian cyber operations in the digital domain

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